HealthConnect — Model Improvement, Error Analysis & Validation (Week 6, Data Science Track)

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Dataset: `HealthConnect_Appointment_Data.csv` (5,000 appointments), using the prepared/engineered version from the Data Analytics Week 5 pipeline.

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1. Where this picks up

No Data Science track baseline existed at the start of Week 6 — the assigned teammates did not deliver Week 5 work. To meet the Week 6 brief's requirement to "analyse the weaknesses of your Week 5 baseline model," a baseline had to be built first, then improved and validated in the same pass. Both stages are documented below so the progression (baseline → error analysis → refinement → improved model → validation) is auditable, matching what Week 6 expects to see.

2. Prediction target and modelling scope

Target: `is_no_show` — 1 if `appointment_outcome == "No-Show"`, 0 if `appointment_outcome == "Attended"`.

Cancelled appointments are excluded from the model. This follows directly from the Data Analytics Week 4 finding that *"Cancelled must stay distinct from No-Show (different failure mode)"* — a cancellation is a known, actioned event; a no-show is an unflagged loss of a slot. Mixing them would blur the thing the clinic actually needs predicted ahead of time: which **unflagged** appointments are at risk of silently not showing.

3. Features used, and why (Data Analytics → Data Science mapping)

Feature	Included?	Data Analytics justification
`booking_lead_days`	Yes	Strongest predictor in Week 5 DA analysis: 27.8% no-show at 0-7 days → 67.7% at 46-60 days
`previous_no_shows`	Yes	Second strongest: 43.5% → 68.8% across the no-show history range
`distance_to_clinic_km`	Yes	Cliff effect at 15+ km (54.1%), flat below — DA found this is a threshold effect, not a smooth gradient

Feature	Included?	Data Analytics justification
`reminder_sent` / `reminder_channel`	Yes (weak signal kept)	DA Week 4 found "only a weak effect — weaker than the business scenario implies," but weak ≠ zero, and it is the one actionable/interventable lever in the set
`previous_appointments`, `waiting_time_minutes`	Yes (control variables)	Included for completeness/context, not flagged as strong by DA, but plausible operational signal
`age`, `gender`, `appointment_type`, `appointment_day`, `appointment_time`	**No**	DA Week 4/5: "Appointment logistics and demographics show no meaningful separation" — excluding these avoids adding noise and avoids encoding demographic bias into a clinical operations model

This table **is** the cross-track integration evidence: every modelling decision traces back to a specific, already-published Data Analytics finding rather than being chosen from scratch.

4. Baseline model

- **Algorithm:** Logistic Regression, `class_weight="balanced"`, standardized features.
- **Rationale:** Per the Week 5 DS brief, the baseline's job is to establish a reliable floor, not the best model — a simple, interpretable linear model is the right choice for that.
- **Split:** 75/25 train/test, stratified on the target.
- **Results (test set, n=1,185, stratified 25% holdout):**

Metric	Value
Accuracy	0.620
Precision	0.620
Recall	0.667
F1	0.642
ROC-AUC	0.665

See `outputs/figures/baseline_confusion_matrix.png`, `baseline_roc_curve.png` and `baseline_coefficients.png`.

An ROC-AUC of 0.665 confirms the three DA-identified variables carry real predictive signal well above chance (0.5), but the model is far from a precise predictor on its own — consistent with the DA track's own framing that these three variables explain meaningful *patterns*, not the full picture of why any single patient does or doesn't show up.

5. Error analysis

`scripts/error_analysis.py` splits the baseline's test-set errors into:

- **False negatives** — predicted Attend, actually No-Show. These are the costlier error for HealthConnect: a slot that looked safe silently fails.
- **False positives** — predicted No-Show, actually Attended. Costly only if the output drives an intervention (e.g. overbooking, a nudge) that then proves unnecessary.

Results: of 1,185 test-set predictions, 735 were correct, 248 were false positives (20.9% of the test set) and 202 were false negatives (17.0% of the test set) — the baseline is somewhat more prone to over-predicting no-shows than missing them.

Error rates are also broken down by `lead_time_band`, `distance_band`, and `prior_no_show_bracket` to check whether the baseline's mistakes cluster in a specific segment — see `outputs/figures/baseline_error_rate_by_*.png` and `outputs/models/baseline_error_analysis.json`.

The error rate is not flat across lead-time bands — it peaks at **15-30 days (45.9% error rate)** and **31-45 days (43.6%)**, and is lowest at the extremes: 0-7 days (27.9%) and 46-60 days (29.8%). This is exactly the non-linear pattern that motivates feature refinement below: the baseline does reasonably well where risk is clearly low (short lead time) or clearly high (long lead time + the double-risk stacking effect), and struggles most in the ambiguous middle bands where a single linear coefficient on lead time can't represent the curve's actual shape.

Finding used to drive refinement: the DA Week 5 confounding check showed lead time and prior no-shows are independent *but stack* — a "double-risk" group (long lead time **and** 2+ prior no-shows) reaches ~73% no-show risk, well above either factor alone. A linear model with independent coefficients cannot represent that multiplicative stacking effect — it will systematically under- or over-predict at the interaction's edges. This is the direct cause investigated in error analysis and the direct target of the feature refinement below.

6. Feature refinement

Three engineered features were added for Week 6, each traceable to a DA finding or an error-analysis result:

1. **`double_risk_flag`** — long lead time (46-60 days) AND 2+ prior no-shows. Encodes the DA-identified ~73%-risk stacking group directly.

2. `leadtime_x_priornoshow` — a raw interaction term, given to the refined Logistic Regression so a linear model can also represent some of the stacking effect.
3. `distance_15km_plus` — DA found the distance effect is a threshold, not a gradient (flat below 15km, a jump at 15km+). A binary flag matches that shape better than the continuous value for a linear model.

7. Improved model and comparison

Two refined variants were trained on the same refined feature set:

- **Refined Logistic Regression** — same model family as baseline, but with the engineered interaction/threshold features, to isolate how much of the improvement comes from features alone vs. model class.
- **Random Forest** (300 trees, `max_depth=8`, `min_samples_leaf=15`, `class_weight="balanced"`) — chosen because tree ensembles learn interactions and non-linear thresholds natively, which is exactly the gap the error analysis identified in the baseline.

All three (baseline, refined LogReg, Random Forest) are compared on accuracy, precision, recall, F1 and ROC-AUC:

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression (baseline features)	0.620	0.620	0.667	0.642	0.665
Logistic Regression (refined features)	0.618	0.619	0.657	0.637	0.665
Random Forest (refined features)	**0.624**	**0.633**	0.632	0.633	**0.666**

See `outputs/models/model_comparison.csv` and `outputs/figures/model_comparison_bars.png` / `model_comparison_roc.png`.

Honest read of these numbers: the improvement is real but small — the Random Forest edges out the baseline on accuracy and precision, and both refined models land within noise of the baseline on ROC-AUC. The engineered interaction/threshold features did **not** produce the large lift the error-analysis hypothesis predicted. The most likely explanation is that `booking_lead_days` and `previous_no_shows` already carry most of their joint signal individually — the three core DA-identified variables are the ceiling on what this feature set can predict, and no amount of feature engineering or model-class change moves far past that ceiling with the

current feature set. This is a legitimate, reportable Week 6 finding in its own right: **the model validates that DA's three variables are doing essentially all the predictive work already identified**, and further gains would need genuinely new information (e.g. patient-level history beyond what's in this dataset, weather, appointment-type-specific patterns) rather than reshaping the existing three.

7.5 Robustness check: did we just confirm what we wanted to see?

Every model up to Section 7 was only ever given features the Data Analytics track had already endorsed. That risks circularity: "different models agree the three DA variables are the ceiling" isn't a strong claim if the models were never shown anything else, and the single train/test split in Section 7 could itself be a lucky or unlucky draw. Two checks were added specifically to test the *approach*, not just to search for a better score (`scripts/robustness_check.py`):

Check 1 — 5-fold stratified cross-validation, restricted vs. full feature set, across three model families (Logistic Regression, Random Forest, Gradient Boosting):

Feature set	Logistic Regression	Random Forest	Gradient Boosting
Restricted (DA-approved only)	**0.682 ± 0.012**	0.677 ± 0.012	0.672 ± 0.012
Full (+ age, gender, appointment_type, day, time)	0.677 ± 0.010	0.675 ± 0.010	0.672 ± 0.009

Two results here matter:

- **Cross-validation raised every restricted-feature metric** versus the single-split numbers in Section 7 (e.g. Logistic Regression 0.682 vs. 0.665) — the original single split understated performance. This is addressed as a limitation below rather than left silently corrected.
- **Adding back the variables DA excluded did not help — it made every model marginally worse.** This is the actual rigorous version of "DA's exclusion is correct": not "DA said so," but "a model given the opportunity to use these variables gets no benefit and picks up a small amount of noise instead." The Section 3 feature table's exclusion decision is now evidence-based, not just inherited.

Check 2 — permutation importance on the full feature set (which directly measures a feature's effect on held-out performance, avoiding the known bias of tree-based impurity importance toward continuous/high-cardinality features): `booking_lead_days` (0.097–0.105) and `leadtime_x_priornoshow` (0.025–0.029) dominate by one to two orders of magnitude. A handful of DA-excluded variables (`appointment_type_Follow-up`, `appointment_day_Thursday`/`Wednesday`, and `age` for Gradient Boosting)

technically rank in the "top 10" of 31 features, but at importance values of 0.0001–0.004 — negligible next to the top two. Ranking 10th among mostly-zero-importance features is not the same as mattering; the precise finding is "statistically indistinguishable from noise," not "zero."

An uncomfortable finding this surfaced, reported rather than dropped: `distance_to_clinic_km` and `distance_15km_plus` show *very low* permutation importance in both models — well below what the DA "15km cliff" narrative (Section 3) would suggest. This is flagged honestly in Limitations (Section 10) rather than smoothed over, since the point of this check was to find exactly this kind of tension if it existed.

Conclusion: the restricted DA-approved feature set is validated, not just assumed — this is the stronger, tested version of Section 7's "three variables are the ceiling" claim. Full results: `outputs/models/robustness_cv_results.csv`, `outputs/models/permutation_importance_*.csv`, `outputs/models/robustness_summary.json`.

Independent cross-confirmation from the DA side: the Data Analytics Week 6 KPI validation (`../week 6/reports/`) bootstrap-resampled each KPI's confidence interval and found the same thing through a completely different method — KPI 3's distance-band confidence intervals overlap heavily across the middle bands, while KPI 1, 2 and 5 stay well-separated. Two unrelated techniques (permutation importance here, bootstrap CIs there) agreeing that distance is the weak link is stronger grounds for the downgrade than either check alone.

8. Candidate model recommendation

Revised from an earlier single-split-based pick. The Section 7 single-split comparison originally favoured Random Forest (0.666 ROC-AUC vs. 0.665 for refined Logistic Regression) — a margin so small it was within one split's sampling noise. The Section 7.5 cross-validation resolves this properly: averaged over 5 folds, **refined Logistic Regression (restricted feature set) wins outright — 0.682 ± 0.012, ahead of Random Forest (0.677 ± 0.012) and Gradient Boosting (0.672 ± 0.012)**.

Candidate: Logistic Regression, refined features (`leadtime_x_priornoshow`, `double_risk_flag`, `distance_15km_plus`), restricted to the DA-approved variable set. This is preferable on more than the metric alone:
- It wins under cross-validation, not a single split.
- It is directly interpretable (readable coefficients), which the business-relevance use case (Section 9) benefits from directly.
- The win traces to the *engineered features*, not to reaching for a more complex model class — a cleaner, more defensible story than "the fancier model happened to edge ahead."

See `outputs/models/candidate_model.json` (updated) and `outputs/models/robustness_cv_results.csv`.

This is not a production deployment decision. Per the Week 6 brief, "Week 6 is still not the final production deployment stage" (explicitly stated for the ML Engineering track, and the same caution applies here) — this candidate is what should be handed to a Machine Learning Engineering track for pipeline integration in a later week, not shipped directly.

9. Business relevance

- A no-show risk score built on these features can flag the double-risk segment (long lead time + prior no-shows) for proactive intervention — e.g. an extra reminder touch or a confirmation call — rather than treating all bookings uniformly.
- Because `reminder_sent`/`reminder_channel` is the one actionable lever in the model, the model's output is most useful paired with a decision rule like: *for patients in the top risk band, escalate reminder channel or add a second touchpoint* — not as a silent scoring exercise.
- The distance threshold (15km+) is a real pattern in the raw data (DA's box-plot/segment analysis) and a plausible target for a non-reminder intervention (transport support, telehealth), **but Section 7.5's permutation-importance check found it contributes very little to the model's actual predictive performance** relative to lead time and prior no-shows. Treat a distance-based intervention as a hypothesis worth a small pilot, not as something the model has confirmed will move the needle — see Limitations.

10. Limitations

- **No real Data Science teammate reviewed this work.** It was built independently by the Data Analytics track intern as a stand-in, using standard classification technique — it has not had a second technical reviewer.
- **Class imbalance handling** relies on `class_weight="balanced"` rather than resampling (SMOTE, etc.) — reasonable at this dataset size and no-show rate (~48.5%, not severely imbalanced), but worth revisiting if the real production population differs.
- **No hyperparameter search** was run for the Random Forest or Gradient Boosting — parameters were chosen conservatively (bounded depth, minimum leaf size / shallow trees) to reduce overfitting risk given the moderate dataset size (5,000 rows), not tuned via cross-validated search.
- **Single train/test split in Sections 4–7** — superseded for the model comparison and candidate selection by the 5-fold cross-validation in Section 7.5, which is now the authoritative comparison. The Section 4–7 single-split numbers are kept in this report for the error-analysis narrative (which depends on having one fixed test set to inspect row-by-row) but should not be read as the final performance estimate.

- **Distance's low permutation importance is unresolved, not explained away.** DA's box-plot evidence for a 15km cliff is real, but the trained models place very little predictive weight on it relative to lead time. Plausible explanations not yet tested: the cliff may be partially redundant with something already captured by lead time or prior no-shows in this synthetic dataset, or the effect may be real but small in absolute contribution once the two dominant variables are accounted for. This should be investigated before any distance-based intervention is presented as model-validated.
- **Synthetic/fictional dataset** — HealthConnect is a fictional clinic scenario; conclusions are illustrative of the *method*, not claims about real-world clinic behaviour.

11. Week 7 testing requirements

For whoever picks this up next (real Data Science teammate, or continued independent work):

1. ~~Cross-validate the candidate model~~ — done in Section 7.5; carry the same 5-fold approach forward for any future model changes.
2. Run a hyperparameter search (grid/random search) on the Logistic Regression's regularization strength, and on Random Forest/Gradient Boosting if either is reconsidered, before treating any metric as final.
3. Test calibration of the predicted probabilities (not just classification metrics) — important if the score is used to rank patients by risk rather than just classify them.
4. Stress-test the double-risk flag's edges (e.g. 45 vs 46 days) to confirm the threshold isn't overfit to this dataset's exact band boundaries.
5. **Investigate the distance permutation-importance gap** (Section 7.5/10) — check for redundancy with lead time/prior no-shows (e.g. partial correlation or a model trained on distance alone vs. distance added last), before recommending a distance-based intervention as model-backed.
6. If a real ML Engineering pipeline exists by Week 7, validate the model loads and predicts correctly through that pipeline's actual input/output contract, not just in this standalone script.

12. Cross-track contribution (evidence)

- **Track collaborated with:** Data Science (simulated — see catch-up note).
- **Project dependency:** Data Analytics Week 4/5 findings → Data Science Week 5/6 model features.
- **Information/output received:** the DA track's own validated findings (this document's author's own Week 4/5 work), since no DS output existed to receive.
- **Information/output provided:** this full DS Track deliverable (baseline model, error analysis, refined/improved model, candidate recommendation) — ready to hand to the Data Analytics Week 6 report as "findings validated by an independent model," and to a future ML

Engineering integration.

- **Integration activity completed:** built a classification model entirely from DA-identified features, with every feature choice explicitly traced back to a specific DA finding (Section 3 table).
- **What changed as a result:** the DA track's qualitative findings ("lead time and prior no-shows stack to ~73% risk") are now backed by a quantitative model that can score any new appointment's no-show risk, not just describe the three historical variables in isolation.
- **Evidence:** this report, `outputs/models/*.json`/`*.csv`, `outputs/figures/*.png`, and the reproducible scripts in `scripts/`.